

NAME = SADASHIV

USN = 2VX24UE044

EXPT = 7(a)

TITLE = 0/1 Knapsack Problem using Dynamic Programming

In [1]:

```
1 n = int(input("Enter the number of items: "))
2
3
4 print("Enter the weight and profit for each item:")
5 weight = []
6 profit = []
7 for i in range(n):
8     w = int(input("Weight of item {}: ".format(i+1)))
9     p = int(input("Profit of item {}: ".format(i+1)))
10    weight.append(w)
11    profit.append(p)
12
13 capacity = int(input("Enter the capacity of the Knapsack: "))
14
15 def DP_KnapSack(capacity, weight, profit, n):
16     K = [[0 for x in range(capacity + 1)] for x in range(n + 1)]
17
18     for i in range(n + 1):
19         for w in range(capacity + 1):
20             if i == 0 or w == 0:
21                 K[i][w] = 0
22             elif weight[i-1] <= w:
23                 K[i][w] = max(profit[i-1] + K[i-1][w-weight[i-1]], K[i-1][w])
24             else:
25                 K[i][w] = K[i-1][w]
26
27     return K[n][capacity]
28
29 max_profit = DP_KnapSack(capacity, weight, profit, n)
30 print("=====")
31 print("Maximum Profit Earned = ", max_profit)
32 print("=====")
```

```
Enter the number of items: 3
Enter the weight and profit for each item:
Weight of item 1: 15
Profit of item 1: 100
Weight of item 2: 10
Profit of item 2: 150
Weight of item 3: 20
Profit of item 3: 120
Enter the capacity of the Knapsack: 50
=====
Maximum Profit Earned = 370
=====
```

In []:

1